Recitation: Rational Functions

Problem 1. Consider the rational function given below:

$$f(x) = \frac{x^2 + 10x + 25}{x^2 + 8x + 15}$$

Complete the following parts:

(a) Find the domain.

(b) Find any y-intercepts.

(c) Find any x-intercepts.

(d) Find any vertical asymptotes.

(e) Find any horizontal asymptotes.

(f) Find any holes.

Problem 2. Consider the rational function given below:

$$g(x) = \frac{2x^3 + x^2 - x}{x^3 + 2x^2 - 3x}$$

Complete the following parts:

(a) Find the domain.

(b) Find any y-intercepts.

(c) Find any x-intercepts.

(d) Find any vertical asymptotes.

(e) Find any horizontal asymptotes.

(f) Find any holes.

Problem 3. For each of the following, find a rational function with the given properties:

(a) A function with domain $(-\infty, 5) \cup (5, \infty)$ and a root at $x = 2$.

(b) A function with a vertical asymptote at $x = 4$ and horizontal asymptote $y = 3$.

(c) A function with no y-intercept.

(d) A function with a hole at $(1, -2)$.

Problem 4. Use the plot below to sketch the graph of $f(x) = \dfrac{x^2 - x - 2}{x^2 - 2x - 3}$.